

The idea of a function

A rule that gives each input exactly one output — as a table, a formula, and a graph.

LESSON 139–140

2 × 40 MIN

MATEMATIKA 7, §60

S8 9.4

LESSON CLOCK

15'

20'

25'

20'

5'

What a function is	15 min
Three ways of giving one	20 min
Values, forwards and backwards	25 min
The domain	20 min
Homework	5 min

LEARNING OBJECTIVES

- State what makes a rule a function.
- Move between a table, a formula and a graph of the same function.
- Find the value of a function at a given argument, and the argument for a given value.
- Name the domain of a simple function.

TEXTBOOK

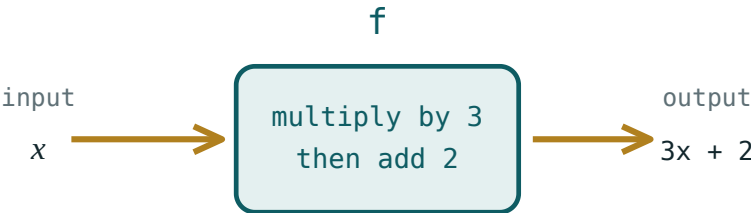
National	pp. 415–422
Cambridge	Stage 8, pp. 90–95
Workbook	Exercise 9.4

01

Explanation

What a function is

A **function** is a rule that assigns to each value of x exactly one value of y . One input, one output — never two.



A function as a machine: one input, one output

RULE	A FUNCTION?	WHY
y is twice x	yes	one answer every time
y is the area of a square of side x	yes	$y = x^2$
y is a number whose square is x	no	$x = 9$
y is the day of the week for a date x	yes	each date has one day

THE VERTICAL-LINE TEST

On a graph, a rule is a function exactly when no vertical line meets the curve twice — because a second meeting would be a second y for the same x .

Three ways of giving one

WAY	EXAMPLE	BEST FOR
a formula	$y = 2x + 1$	calculating any value exactly
a table	a row of x and a row of y	seeing a few values at once
a graph	points $(x; y)$ plotted	seeing the shape of the change
words	“double it and add one”	stating the rule to somebody

x	-2	-1	0	1	2	3
$y = 2x + 1$	-3	-1	1	3	5	7
$y = x^2$	4	1	0	1	4	9

ALL THREE DESCRIBE ONE OBJECT

The table is a sample of the formula; the graph is the whole table drawn at once. Being able to move between them is the skill this lesson is really about.

Values, forwards and backwards

Write $f(x)$ for the value at x . Finding $f(3)$ is substitution; finding x from a known $f(x)$ is solving an equation.

QUESTION	WORKING	ANSWER
$f(x) = 2x + 1, f(3)$	$2 \cdot 3 + 1$	7
$f(x) = 2x + 1, f(-4)$	$-8 + 1$	-7
$f(x) = 2x + 1 = 11$	$2x = 10$	$x = 5$
$f(x) = x^2, f(x) = 25$	$x^2 = 25$	$x = \pm 5$

ONE Y MAY COME FROM TWO XS

A function forbids two outputs for one input, not two inputs for one output. $f(x) = x^2$ sends both 5 and -5 to 25 and is still perfectly a function.

The domain

The **domain** is the set of values of x for which the rule gives an answer.

FUNCTION	DOMAIN	WHY
$y = 2x + 1$	every number	nothing can go wrong
$y = \frac{1}{x}$	$x \neq 0$	division by zero
$y = \frac{1}{x - 3}$	$x \neq 3$	the denominator vanishes
the area of a square of side x	$x > 0$	a side cannot be negative

TWO KINDS OF RESTRICTION

One comes from the algebra — a zero denominator; the other from the situation — a length must be positive. Both are part of the answer when a function describes something real.

02 Worked examples**EXAMPLE 1** For $f(x) = 2x + 1$, find $f(3)$ and $f(-4)$.

① $f(3) = 2 \cdot 3 + 1 = 7$

② $f(-4) = 2 \cdot (-4) + 1$

③ $= -8 + 1 = -7$

ANSWER

7 and -7

EXAMPLE 2 For $f(x) = 2x + 1$, find x when $f(x) = 11$.

① $2x + 1 = 11$
Now it is an equation.

② $2x = 10$

③ $x = 5$

ANSWER $x = 5$

EXAMPLE 3 Is “ y is a number whose square is x ” a function?

- ① Take $x = 9$.
- ② Both 3 and -3 qualify.
- ③ One input gives two outputs.
- ④ So it is not a function.

ANSWER

No

03 Interactive model

Draw a function machine on the board and feed it numbers the class calls out; the definition arrives as a description of what everyone has just watched.

Feed a value into $\frac{1}{x-3}$

$$f(x) = \frac{1}{x-3}$$

x 5

denominator 2

value 0.5

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Function	Funksiya	Функция
Argument	Argument	Аргумент
Value of a function	Funksiya qiymati	Значение функции
Independent variable	Erkli o'zgaruvchi	Независимая переменная
Dependent variable	Bog'liq o'zgaruvchi	Зависимая переменная
Domain	Aniqlanish sohasi	Область определения
Table of values	Qiymatlar jadvali	Таблица значений
Graph	Grafik	График

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A function gives each input:

many outputs

exactly one output

no output

two outputs

The vertical-line test fails when a vertical line:

misses the curve

touches it once

meets it twice

is an axis

For $f(x) = 2x + 1$, $f(3)$ is:

5

7

6

9

$f(x) = 11$ gives x equal to:

The domain of $y = \underline{1}x - 3$ is:

$f(x) = x^2$ sends both 5 and -5 to 25, so it is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $f(x) = 2x + 1: f(0)$

ANSWER 1

2. $f(3)$

ANSWER 7

3. $f(-4)$

ANSWER -7

4. $f(x) = x^2: f(3)$

ANSWER 9

5. $f(-3)$

ANSWER 9

6. The domain of $y = 2x + 1$

ANSWER Every number

7. The domain of $y = \frac{1}{x}$

ANSWER $x \neq 0$

Medium · the standard question

7 PROBLEMS

1. $f(x) = 2x + 1 = 11$: find x

ANSWER $x = 5$

2. $f(x) = x^2 = 25$: find x

ANSWER $x = \pm 5$

3. The domain of $y = \frac{1}{x-3}$

ANSWER $x \neq 3$

4. Is “ y^2 equals x ” a function of x ?

ANSWER No

5. Is the area of a square a function of its side?

ANSWER Yes

6. A table for $y = 2x + 1$ at $x = -2, 0, 2$

ANSWER $-3, 1, 5$

7. The vertical-line test decides

ANSWER whether a graph shows a function

Hard · stretch and reason

7 PROBLEMS

1. $f(x) = 3x - 2$: find $f(f(1))$

ANSWER 1

2. $f(x) = x^2 - 4$: find every x with $f(x) = 0$

ANSWER $x = \pm 2$

3. The domain of $y = \frac{1}{x^2 - 4}$

ANSWER $x \neq \pm 2$

4. A function whose graph is a horizontal line

ANSWER $y = c$

5. Why is a vertical line not the graph of a function?

ANSWER One x would have every y

6. $f(x) = 2x + 1$: for which x is $f(x) = x$?

ANSWER $x = -1$

7. The domain of the perimeter of a square as a function of its side

ANSWER $x > 0$

07 Homework

Homework — 5 tasks

Say which is the argument and which is the value in every task.

- For $f(x) = 3x - 5$, find $f(0)$, $f(2)$ and $f(-1)$.
- For the same function, find x when $f(x) = 7$.
- Make a table of $y = x^2 - 1$ for $x = -3$ to 3 .
- Give the domain of $y = \frac{1}{x + 2}$.

5. Explain why “ y is a number whose square is x ” is not a function.